

Industrial Assignment (week-5)**Statistic and linear algebra**

1. Consider the Iris dataset and perform an analysis of the statistical summary of the dataset

The statistical summary of Iris dataset is as follows.

	sepal-length	sepal-width	petal-length	petal-width
1				
2	count	150.000000	150.000000	150.000000
3	mean	5.843333	3.054000	3.758667
4	std	0.828066	0.433594	1.764420
5	min	4.300000	2.000000	1.000000
6	25%	5.100000	2.800000	1.600000
7	50%	5.800000	3.000000	4.350000
8	75%	6.400000	3.300000	5.100000
9	max	7.900000	4.400000	6.900000

Analyse and explain statistical metrics from above summary.

2. Explain Central tendency, standard deviation with respect to dataset.
3. Write a note on Eigen values and Eigen vectors. Explain Eigen values and eigen vectors with simple 2*2 matrices.
4. Explain the different data transformation approaches and illustrate with a Python program.
5. Write a program to remove data duplication.